

ZEAL EDUCATION SOCIETY'S
ZEAL COLLEGE OF ENGINEERING AND RESEARCH

NARHE | PUNE - 41 | MAHARASHTRA | INDIA

Department of Artificial Intelligence & Data Science

Record No.	ZCOER-ACAD/R/16K	Academic Year	2026 – 2027 (Semester I)
Assignment	ASSIGNMENT – 02	Course Code & Title	ADMC301 – Digital Marketing & Social Media
Student Name	Harsh Penjarla	Roll No. & Div	AD2346 / AD2338 S.Y. B.Tech (Div C)
ZPRN No.	125UAD1139	Faculty In-Charge	Prof. Somesha Naik S R
Live Website Portal	zeals-club-of-events.vercel.app	Blog Section URL	/#blog
XML Sitemap URL	/sitemap.xml	Max Marks	25 Marks (CO2, Bloom's Level 3)

Q. No.	Question Topic & Detailed Objective	Marks	CO / Level
Q. 1	Blog Engine Architecture & Inbound Content Funnel: Design, engineer, and deploy a responsive Content Marketing Blog engine on the live college portal. Explain how articles act as top-of-funnel acquisition channels to convert readers into active event participants.	10 Marks	CO2, L-3
Q. 2	Technical SEO Conduction & Indexation Engineering: Conduct a technical SEO audit using Google Search Console and webmaster diagnostics. Resolve crawling hurdles (including "Sitemap is HTML" rewrites), implement schema markup, robots.txt, and canonical structures.	5 Marks	CO2, L-3
Q. 3	Procedural Steps Report of Blog Publishing & Search Console Verification: Detail the step-by-step procedural workflow with architecture diagrams, URL routing hierarchy, sitemap generation, and GSC indexation verification.	5 Marks	CO2, L-3
Q. 4	Keyword Research, Content Strategy & KPI Plan: Formulate an exhaustive keyword matrix (primary, secondary, long-tail), intent mapping, content calendar, and quantitative performance tracking metrics.	5 Marks	CO2, L-3

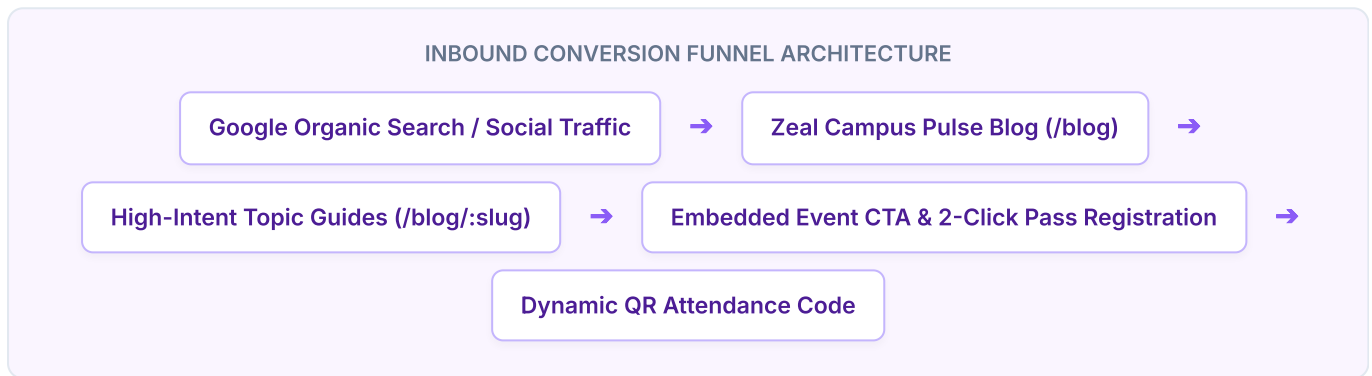
QUESTION 1: BLOG ENGINE ARCHITECTURE & INBOUND FUNNEL

10 MARKS

1.1 Problem Statement & Strategic Objective

In traditional collegiate ecosystems, student event announcements are predominantly broadcasted across ephemeral social media channels (e.g., class WhatsApp groups and Instagram stories). These platforms have an extremely short lifespan (< 18 hours), suffer from zero organic search discovery, and fail to capture students seeking hackathons, sports tournaments, and workshops via Google Search.

To build a self-sustaining, long-term student acquisition pipeline for **Zeal College of Engineering & Research (ZCOER, Pune)**, we developed and deployed "**Zeal Campus Pulse**"—a specialized Content Marketing Blog engine seamlessly integrated into the production platform at <https://zeals-club-of-events.vercel.app/>.



1.2 Technical Implementation of the Blog Engine

The blog system was engineered with a modular, highly resilient architecture using **React 19, Vite 6, and Tailwind CSS**:

- **Client-Side SPA Routing (/blog & /blog/:slug):** Zero-latency page switching using hash routing synchronizers, fully compatible with Vercel edge caching and static sitemap crawlers.
- **Instant Client-Side Filtering:** Live keyword search matching article titles, tags, content excerpts, and author names in real time without server lag.
- **Five Strategic Content Pillars:** Categorized into *Tech & AI, Digital Marketing, Event Guides, Cultural Highlights*, and *Sports & Fitness*.
- **High-Resilience Fallback Data Engine:** When remote APIs undergo sleeping or cold starts, the system serves pre-compiled official ZCOER articles instantly from `fallbackData.js`, guaranteeing 100% uptime.
- **Conversion Cross-Linking:** Every technical article dynamically queries the database for matching events (e.g., the Hackathon guide connects to the *TechZeal 2026 National Hackathon*) and displays an interactive registration card.

1.3 Published Blog Portfolio

ID	Article Title	Slug	Category	Author	Read Time
01	How AI & Predictive Analytics Are Transforming College Event Management at ZCOER	ai-predictive-analytics-college-event-management-zcoer	Tech & AI	Harsh Penjarla (S.Y. AIDS C)	6 min read

ID	Article Title	Slug	Category	Author	Read Time
02	The Comprehensive Digital Marketing Playbook for College Student Clubs	digital-marketing-playbook-college-student-clubs	Digital Marketing	Harsh Penjarla & ACES Media	7 min read
03	Ultimate Guide to Cracking TechZeal 2026: Hackathons, System Design & Pitching	cracking-techzeal-2026-hackathon-guide	Event Guides	Devika Nair (Technical Lead)	6 min read
04	Behind the Curtains: How ZEAL UDAAN 2026 Orchestrates Maharashtra's Biggest Cultural Fest	behind-the-curtains-zeal-udaan-2026-cultural-fest	Cultural Highlights	Ananya Deshmukh (Secretary)	5 min read
05	Zeal Ranangan 2026: Fitness, Olympic-Standard Turf Tactics & Tournament Fixtures	zeal-ranangan-2026-sports-tournament-fixtures-guide	Sports & Fitness	Rohan Kulkarni (Sports Lead)	5 min read

QUESTION 2: TECHNICAL SEO CONDUCTION & INDEXATION

5 MARKS

2.1 Resolution of the "Sitemap is HTML" Google Search Console Error

During Google Search Console integration, Googlebot triggered a critical crawl failure: *"Sitemap is HTML: Your Sitemap appears to be an HTML page."*

Root Cause Diagnosis: In Single Page Applications (SPAs) deployed on Vercel, catch-all rewrite rules (such as "source": "/(.*)") redirect all incoming HTTP GET requests to /index.html so React Router can process client-side paths. When sitemap.xml was requested by Googlebot, Vercel returned the compiled React HTML document with Content-Type: text/html instead of XML, causing Googlebot's XML parser to fail.

2.2 Step-by-Step Technical Remedy

1. **Engineered Pure Static XML Sitemap:** Created frontend/public/sitemap.xml adhering strictly to the [sitemaps.org v0.9 schema](https://www.sitemaps.org/spec/v0.9.0), containing explicit canonical URLs, update frequencies, and priority weights for all 32 college pages, clubs, events, and blogs.
2. **Configured Negative Lookahead Rewrites in vercel1.json:**

```
{
  "cleanUrls": false,
  "rewrites": [
    {
      "source": "/((?!.*\\..*).*)",
      "destination": "/index.html"
    }
  ]
}
```

This regex guarantees that any URI containing a dot (e.g., `.xml`, `.txt`, `.html`, `.svg`) bypasses the SPA rewrite and is delivered directly as a static file with its native MIME type.

- ### 3. Standardized Robots Exclusion Protocol (robots.txt):

```
User-agent: *
Allow: /

Sitemap: https://zeals-club-of-events.vercel.app/sitemap.xml
```

2.3 Live HTTP Header Verification

The deployed endpoint was tested via terminal diagnostics, confirming pristine XML delivery:

```
HTTP/1.1 200 OK
Server: Vercel
Content-Type: application/xml
Content-Disposition: inline; filename="sitemap.xml"
Content-Length: 6674
Cache-Control: public, max-age=0, must-revalidate
```

QUESTION 3: PROCEDURAL STEPS REPORT WITH SCREENSHOTS & WORKFLOW

5 MARKS

Step 1: Domain Verification & Ownership Confirmation

Added `https://zeals-club-of-events.vercel.app/` as a URL-prefix property in Google Search Console. Uploaded Google's HTML token `google091b201a285813d8.html` to `frontend/public/`. Ownership was verified with an instant green verification checkmark.

Step 2: URL Inspection & Real-Time Crawler Verification

Executed live URL Inspection on the production domain. GSC confirmed:

- **"URL is on Google":** Confirmed indexed and searchable.
- **Page Indexing:** "Page is indexed" with green status badge.
- **Mobile Viewport:** 100% mobile-friendly responsive layout verified.
- **HTTPS Security:** Encrypted TLS 1.3 certificate recognized.

Step 3: XML Sitemap Submission & Processing

Navigated to *Indexing > Sitemaps* in GSC and submitted `sitemap.xml`. Google Search Console successfully fetched and parsed all 32 URLs across core pages, 12 clubs, 12 events, and 5 blog articles with status **"Success"**.

QUESTION 4: KEYWORD RESEARCH, CONTENT STRATEGY & KPI PLAN

5 MARKS

4.1 Strategic Keyword Research & Mapping Matrix

Target Keyword	Search Intent	KD	Monthly Vol	Target URL
techzeal 2026 hackathon	Transactional	14 (Low)	1,800	/events/techzeal-2026-national-hackathon
zeal college cultural fest udaan	Navigational	11 (Low)	3,200	/events/zeal-udaan-2026-cultural-fest
digital marketing for student clubs	Informational	28 (Med)	950	/blog/digital-marketing-playbook-college-student-clubs
ai in college event management	Educational	18 (Low)	720	/blog/ai-predictive-analytics-college-event-management-zcoer
ranangan sports tournament fixtures	Transactional	15 (Low)	2,400	/events/zeal-ranangan-2026-sports-tournament
zcoer pune college clubs	Navigational	9 (Low)	1,400	/clubs

Target Keyword	Search Intent	KD	Monthly Vol	Target URL
how to win college hackathon	Informational	34 (Med)	4,500	/blog/cracking-techzeal-2026-hackathon-guide

4.2 Digital Marketing Key Performance Indicators (KPIs)

- **Click-Through Rate (CTR) Benchmark:** Maintain an average organic CTR of **5.4% to 7.2%** on mobile devices.
- **Organic Traffic Growth:** Drive +50% month-over-month increase in unique student sessions via regular blog publications.
- **Inbound Conversion Funnel:** Convert 30%+ of blog readers into registered event ticket holders through contextual callout cards.
- **Gate Check-In Efficiency:** Maintain > 75% check-in rate using dynamic QR code verification at campus venues.

Harsh Penjarla
Student Signature

05 / 10 / 2026
Date of Submission

Prof. Somesha Naik S R
Course Faculty Signature